

Instagram and YouTube bloggers promote it, why should I buy? How credibility and parasocial interaction influence purchase intentions

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Abstract

Nowadays social network influencers play an important role in marketing by introducing products to their audience. In this article, we investigate the persuasion cues related to beauty and fashion influencers present on YouTube and Instagram. More precisely, we investigate how the para-social interaction (PSI) the audience creates with the online influencer, along with their perceived credibility, are related to the purchase intention and how they are, in turn, related to the social and physical attractiveness and attitude homophily. We base our research on four beauty influencers popular in France and control our results by the age of the participants and by the influencer. We find that attitude homophily is positively related to PSI but, surprisingly, the physical attractiveness shows negative relationship or no evidence of relationship. Both credibility of the influencers and PSI exhibit significant and positive relationships to purchase intention.

Keywords: Parasocial interaction, Credibility, Purchase Intention, Social networks, Influencer, Beauty Brands, Persuasion, Communicator

1. Introduction

Recently, there has been a spate of interest in online social media, such as Facebook, Twitter, YouTube, Instagram and others (Knoll, 2016; Dwivedi et al., 2018; Shiau et al., 2017). Presence on social networks has become a crucial part of marketing strategies: the promotion of products online via business accounts and advertising on social networking platforms has proven to be efficient (De Vries and Carlson, 2014; Gensler et al., 2013).

In addition to the branded posts and fan pages found on social media, users actively generate and publish multimedia content including their opinions on brands and products. Such content, also known as User Generated Content, has proven to be more popular and effective than professional advertisements (Welbourne and Grant, 2016; Aral et al., 2013; Lipizzi et al., 2015). The traditional

word-of-mouth (WOM) has moved online becoming an electronic word-of-mouth (eWOM) (Chang et al., 2015; Hennig-Thurau et al., 2004; Alalwan et al., 2017; Kapoor et al., 2018). Thus, brands and customers engage in collaborative processes of knowledge exchange and product-related information sharing (Merz et al., 2018; Shi et al., 2016).

Certain users can be active creators of online content: personal stories, ideas, reviews, opinions, feelings, emotions, etc., intended for a given audience (Chau and Xu, 2012; Hsu et al., 2013). Bloggers can become opinion leaders and their posts can impact brands and products and influence potential customers (Smith et al., 2007). Therefore, we refer to them as influencers (Susarla et al., 2016; Zhu et al., 2015; Uzunolu and Kip, 2014).

Influencers are present in all sectors: health and fitness, fashion and beauty, food, high-tech and other (Raggatt et al., 2018; Klassen et al., 2018). YouTube, Instagram and Facebook bloggers often present the products they have tested providing their opinion or promoting them online to other users. The posts of influencers on Instagram usually

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take the form of an image or video enriched with embedded content and a textual description. The popularity of bloggers and their UGC has brought a new form of marketing called 'influencer marketing' where brands collaborate with influencers to promote their products (De Veirman et al., 2017).

Brands and businesses are increasingly interested in finding ways to leverage 'popular figures' as brand ambassadors on social media, and those are not necessarily famous singers, actors or athletes. As recently reported by an industrial study conducted by Business Insider (Gallagher, 2018), it is critical for many brands to find and manage relationships with social media influencers. According to the same study, investments in this area are poised to reach between \$5 billion and \$10 billion in 2022 (that is a five-year compound annual growth rate of 38%), especially in the sectors of fashion, beauty, parenting and tourism. Whereas, many other industrial studies are reporting similar results (Statista, 2018), research in what we call now 'influencer marketing' is still limited (De Veirman et al., 2017; Aral, 2011).

An important question arises: **what makes a blogger, or any other content creator, influence their followers?** Businesses need to understand the 'influencer' phenomenon to make the right investment in influencer marketing and influencers. Also, the influencers need insights to be more persuasive while working with brands and products and promoting the products to their followers (Hollebeek, 2011; Hollebeek et al., 2014; Aral, 2011).

Many researchers have focused on celebrity endorsers (Ford, 2018; McCormick, 2016). But Social influencers are neither traditional celebrities, like famous actors or athletes, nor are their posts the traditional advertisements (Abidin, 2016). It was shown that bloggers with high audiences are more likable (De Veirman et al., 2017) and their 'promoting' posts are better appreciated by potential customers than traditional ads (de Vries et al., 2012) even though the quality of the content looks less professional (Hautz et al., 2014) or when the content is marked as 'sponsored' (Müller et al., 2018). Susarla et al. (2016) have noticed the ephemeral nature of attention and inequality in the popularity of user-generated content. However, more knowledge and cues about the persuasiveness of YouTube and Instagram influencers are yet to be discovered.

The aim of this article is to fill in this gap by enhancing the comprehension of the persuasion cues

related to social media influencers and their impacts on the attitudinal effects of influence attempts. We focus on Instagram and YouTube bloggers of the beauty and fashion sector because of the intensive use of influencer marketing in this sector and within these two social network platforms (Garland and Reed, 2018; FashionMonitor, 2016).

Building on the persuasion theory developed by Petty and Cacioppo (1986), previous researchers found the importance of the arguments in the persuasion process, but, according to Mosler (2006), the peripheral cues could play a more important role than the arguments themselves in persuading a group. Users connect to social media websites to fulfill their information, entertainment, and mood management needs (Shao, 2009). The content is entertaining and easy to absorb, adapted to the young population that refers to the peripheral route of the Elaboration likelihood model (ELM) (Petty and Cacioppo, 1986). Although, the physical attractiveness of a beauty and fashion blogger can also be viewed as an argument for this particular industry.

In this article, we focus on persuasion factors related to the beauty and fashion influencers. We study how the credibility of the blogger, para-social interaction, social and physical attractiveness and homophily attitude are related and affect the purchase intention.

We base our study on an original survey-based dataset related to four highly followed beauty and fashion bloggers in France: EnjoyPhoenix, DazzlingDrew, Sananas2016 and Georgia Secret (Horackova), who are all present and active on YouTube and Instagram. The focus on women is motivated by the sector studied, in which this category of customers is a privileged target. Age differences will also be investigated, as age has been recognized as a significant explanatory factor of the interaction mechanisms within social media and their impacts in marketing (Goodrich, 2013; Muk, 2013).

This paper is structured as follows. Section two presents the literature review related to persuasion and para-social interaction theories. Section three presents the research model and the hypotheses studied in this work. Section four describes the methodology used in this article: questionnaire development and administration as well as statistical procedure. Section five highlights the results of the global sample and multi-group analyses. Section six proposes a discussion of the results and outlines the theoretical and practical implications of this work.

Section seven concludes the article and outlines limitations and also future avenues to which this study opens the paths.

2. Theoretical background

2.1. Communicators and persuasion

While studying persuasion, researchers often refer to the elaboration likelihood model of persuasion (ELM) (Petty and Cacioppo, 1986), where the motivation and abilities of the person processing the message would affect whether the argument quality or other peripheral cues would influence the attitudes, intentions or beliefs of the receiver. The information would be processed using a central route with cognitive responses related to information, facts and arguments or using a peripheral route using peripheral cues: cognitive shortcuts that are easy to process. Such cues could be related to the message and its presentation, to the source of the message or the speaker, the environment, etc.

The communicator and the way he/she is perceived can have an impact on the persuasion of an audience. Three characteristics of a speaker are considered fundamental in persuasion: authority, credibility and social attractiveness, these are related to compliance, internalization and identification, respectively (Kelman, 1958).

Authority is frequently related to the position of the communicator in the social structure. The communicator can have the ability to inflict punishment or give a reward or approval to the message receiver who complies in order to avoid punishment or to obtain a reward (Kelman, 1958).

Credibility could be defined as the degree of the trustworthiness and reliability of the source (Rogers and Bhowmik, 1970). Credibility is related to the perception of the quality of the communication by the audience member. Credibility influences the audience through the internalization process: members share similar values and attitudes with the speaker (Kelman, 1958). Expertise - knowledge and experience the person has in the given domain - is one of the main factors of credibility, along with trustworthiness, and goodwill (Hovland and Weiss, 1951; McCroskey and Teven, 1999). Trustworthiness relates to the perceived honesty of the speaker and the goodwill reflects his/her perceived caring about his/her audience. Generally, if the source is considered being trustworthy, attractive and perceived as an expert, it can influence the attitude and

behavior of the audience (Ohanian, 1990; Petty and Wegener, 1998) including purchase intention (Gunawan and Huarng, 2015).

On online social media, Lin et al. (2016) found that authority, identity, and bandwagon cues explain the perception of credibility of messages on Twitter. Credibility is also found to affect the content people choose to view, or to ignore, on social media, through what Johnson and Kaye (2013) call selective exposure. Credibility were found to be related to the advertising value, and the advertisement coming from the peer on social media is seen to be more credible comparing to traditional media Shareef et al. (2019). The audience in number of followers was found to be positively related to the credibility of the source on Twitter (Jin and Phua, 2014) and Instagram (De Veirman et al., 2017). The credibility of blogs was also found to affect positively the attitude towards the blog (Colton, 2018).

Social attractiveness refers to the likability of the speaker. Attractive speakers can change the attitudes of the audience (Sharma, 1990). Regarding the same arguments used in a message, the attractive speaker is more likely to influence the audience than less appealing ones (Chaiken, 1979). Perceived social attractiveness can be a strong peripheral cue for decision making.

According to Kelman (1958), attractive speakers influence the audience through the identification process. An audience member would feel similar or would like to be similar to the speaker and to create a positive relationship with him/her. Millennials - deemed to be extensive online social network users - frequently identify with celebrities and borrow some aspects of their personalities and lifestyle in order to look like them (Peter, 2004; Boon, 2001). Thus, celebrities and online influencers launch fashion and other trends that are followed by those who admire them. The impact of an action of a celebrity can be even greater when the receivers perceive him/her as someone on whom they can personally rely (McCracken, 1989).

Similarity was also found to be a persuasion facilitating factor (Brock, 1965), particularly if the similarity is relevant to the message (Berscheid, 1966). Similar people can inspire trust as the proposed solutions could also work for the audience due to the similarities between the speaker and the audience member (Levine and Valle, 1975). Social cognitive theory suggests that people are more easily influenced by social figures that are perceived

to be similar to them (Bandura, 1994). For instance, millennials are more likely to buy a product endorsed by someone matching their self-image, according to Sukhdial et al. (2002). Although, the similarity could have an opposite effect if the audience perceives a person as a non-expert (Goethals and Nelson, 1973), particularly concerning factual matters.

The similarity between interacting individuals in terms of beliefs, education and social status is called attitude homophily (Prisbell and Andersen, 1980). This construct is related to the number of interactions the individuals have, as similar communicators are more likely to interact with each other. The repeated interactions help to develop a relationship or, in the case of a celebrity, para-social relationship that is strongly related to the identification process (Brown and Basil, 2010).

2.2. Influencers and para-social interaction

The theory of para-social interaction (PSI) defines the relationship between a spectator and a performer (Horton and Wohl, 1956) with an illusion of intimacy as for the 'real' interpersonal relationships (Dibble et al., 2015). Such a relationship is self-established and the other person could be unaware of the relationship and influence it (Kelman, 1958).

Specifically, women having a para-social relationship with a celebrity express the desire to have the same appearance and belong to the 'club' (Greenwood et al., 2008). In some ways, this could be similar to the relationship created between social media users and online influencers. Online social networking users can create such a relationship with bloggers by subscribing to their channels or blogs and following their posts published on social media. Multiple followers can form an online community where the members share with the blogger similar values, beliefs and interests (Nambisan and Watt, 2011). YouTube channels having one communicator are found to be more popular and influential than channels having multiple speakers (Welbourne and Grant, 2016), which could be due to para-social relationships created with the unique speaker. Celebrities that are able to connect with their audience are more effective in terms of persuasion (Sukhdial et al., 2002) unlike unfamiliar celebrities (McCormick, 2016).

The relationship between an influencer on Instagram, YouTube or Facebook is not unidirectional, as users are able to add comments and discuss the

content, and the content owner has the possibility of replying to the messages and comments related to that content. However, influencers cannot actually respond to all of their followers' requests and engage fully in a true discussion due to a very high number of followers and reactions they might have on their blogs. Accordingly, online bloggers are similar to celebrities on traditional media. Therefore, a fully bidirectional relationship and a 'real' friendship is more illusory than real between a blogger and a follower, as originally put forward by the PSI definition in the context of traditional media celebrities (Labrecque, 2014; Lee and Watkins, 2016). Kim et al. (2015) found that social networking sites usage is positively related to the development of a para-social relationship with celebrities.

For instance, young women follow both celebrities and bloggers on Instagram, both being influential on their followers. However, digital personalities seem to have stronger persuasiveness and credibility as followers are more able to relate to them (Djafarova and Rushworth, 2017). The most appreciated profiles frequently support their followers on Instagram, provide positive reviews and help them by reflecting a positive image of themselves.

Lee and Watkins (2016) demonstrated that para-social interaction between YouTube bloggers and their followers has a positive impact on luxury brand perception, and that physical attractiveness and the perceived resemblance to the blogger is positively related to the para-social interaction. The feedback function available on YouTube does not influence the para-social interaction, according to (Rihl and Wegener, 2017). Hwang and Zhang (2018) conducted a study in China and found that PSI is positively related to purchase intention and influenced by empathy and the low self-esteem of the follower.

3. Research model development

Building on our literature review, we have developed an original research model referring to the fundamental process of internalization and identification applied to Instagram and YouTube influencers. Internalization is related to the credibility of the speaker, while identification is more an affection-related and emotional process that is strongly linked to para-social interaction (Kelman, 1958; Horton and Wohl, 1956; Brown and Basil, 2010).

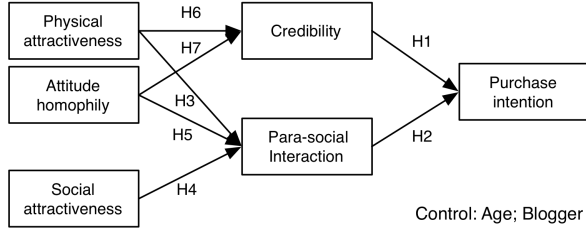


Figure 1: Research Model

We do not consider compliance in our model because influencers on social media are not direct authorities to their followers in the social structure (Gass, 2015). Even if an influencer can be perceived as endorsing a superior social status, a blogger cannot directly reward or punish the follower for purchasing a featured product or not doing so.

In our model, we aim to test the impact of credibility of the beauty influencer and para-social interaction the influencer is able to generate on the intention of their followers to purchase the product featured on the channel. The follower could buy the product because the influencer is perceived to be reliable or believable as formulated by Wathen and Burkell (2002). Following previous works (Fan and Miao, 2012; Sheeraz et al., 2012; Hui, 2017), we formulate our first research hypothesis:

H1: The influencer’s credibility is positively related to the user intention to purchase a featured product.

The follower might also feel a proximity with the influencer and would buy a product because of this perceived proximity expressed and conceptualized, as we have seen in the previous section, in terms of para-social interaction which is reported in the literature as an antecedent of the purchase intention (Kim et al., 2015; Lee and Watkins, 2016; Hwang and Zhang, 2018). Thus, our second research hypothesis is the following:

H2: Para-social interaction between the influencer and his/her followers is positively related to the user intention to purchase a featured product.

Concerning the determinants of para-social interaction, Giles (2002) has reminded that this concept is anchored in an asymmetric relationship between, what he calls media users (social networking sites’ users) and media figures (such as social media influencers). This relationship is built depending on a certain number of factors including how media users perceive media figures as attractive and similar to

them. Attractiveness could be depicted as physical or social attractiveness (Rubin and McHugh, 1987; Rubin and Step, 2000).

In the social psychology literature, physical attractiveness describes how appealing or pleasing a person’s physical features and aesthetic beauty are. These could vary across cultures and periods, but remain strongly related to social interaction (Reis et al., 1980). In recent work and in the context of influence online relationships, physical attractiveness is reported as a predictor of PSI (Lee and Watkins, 2016). Therefore, we hypothesize:

H3: Para-social interaction between the influencer and his/her audience is positively related to his/her physical attractiveness.

Social attractiveness is defined as the likelihood to befriend or to choose the media figure as a social or a work partner. Along with physical attractiveness, social attractiveness has also been demonstrated as a predictor of PSI concerning traditional (television, newspapers) or new media (social media) (Rubin and McHugh, 1987; Lee and Watkins, 2016; Kurtin et al., 2018). Thus, we define our fourth hypothesis:

H4: Para-social interaction between the influencer and his/her audience is positively related to his/her social attractiveness.

Attitude homophily, defined as the degree to which people tend to bond with similar others, is deemed an explaining factor of interpersonal communications and influence-based relationship patterns, especially in social networking studies (McPherson et al., 2001; Aral and Walker, 2012; Walker, 2014). Based on previous works of (Giles, 2002; Eyal and Rubin, 2003; Lee and Watkins, 2016), it is also posited as a determinant of PSI on social media. The next hypothesis we propose to test is:

H5: Para-social interaction between the influencer and his/her audience is positively related to his/her attitude homophily.

As for credibility, the literature describes it as a less subjective or emotional-based determinant of social influence (Gass, 2015). It includes, according to Giffin (1967), expertness, reliability, intentions, activeness and personal attractiveness. In our study related to the beauty and fashion industry, we propose to positively relate credibility to physical attractiveness where it can serve as a selling point. For instance, influencers can show, using their own appearance, how the results expected from the advocated products are confirmed. So that, the lack

of physical appeal and fashion expertise could reduce the perceived credibility of the influencer. We therefore hypothesize:

H6: The credibility of the influencer is positively related to his/her physical attractiveness.

We consider the attitude homophily as a determinant factor of credibility as it refers to the perception of similarities between an influencer and a follower in terms of values. The internalization process refers indeed to similar values that are found in another person. Therefore, we argue that attitude homophily would also be positively related to the credibility of the influencer (Djafarova and Rushworth, 2017).

H7: The credibility of the influencer is positively related to his/her attitude homophily.

Accordingly, our theoretical model includes six constructs and seven research hypotheses that are depicted in Figure 1. It stipulates that credibility and PSI positively affect intention to purchase (respectively, hypotheses H1, and H2); physical attractiveness, social attractiveness and attitude homophily positively affect PSI (respectively, hypotheses H3, H4 and H5). Whereas credibility is positively related to physical attractiveness (H6) and attitude homophily (H7).

As age has been revealed in the literature as a relevant variable to differentiate influence patterns between generations (Hui, 2017; Merz et al., 2018), we also propose to test our model for different user age categories, trying to assess the generational effect within the joint impact of PSI and credibility on the purchase intention. Besides age, the second control variable used in our study is the influencer: we will test our model for the whole sample and also for each of the four influencers studied. Gender is not considered as the whole population under study is female.

4. Methodology

4.1. Questionnaire development and administration

To operationalize our constructs, all the measures used in this study were adapted from the literature. Para-social interaction has been operationalized using Lee and Watkins (2016), attitude homophily using Eyal and Rubin (2003), social and physical attractiveness with reference to Ohanian (1990). For the credibility construct, we adapted the measurement of McCroskey and Teven (1999), by reducing

the number of questions to improve the response rate of our study. Intention to purchase has been operationalized following See-To and Ho (2014). In conformance with the literature, all the items are reflexive and assessed with a 5-point Likert scale. Table 2 summarises the questionnaire.

We collected data using an online survey administered within a large population of users of social network platforms in France who are interested in beauty and fashion content. To construct a more homogeneous sample we selected, using filtering questions, respondents who are familiar with the following influencers: EnjoyPhoenix, Sananas, GeorgiaSecret (a.k.a. Georgia Horackova) and Dazzlingdrew.

All of these four influencers are among the most popular beauty and fashion influencers in France on YouTube and Instagram. Georgia Horackova's blogs also cover fitness and vegan diet in addition to beauty and fashion. All of these four influencers are females living in France (between 20 and 35 years old): DazzlingDrew and EnjoyPhenix are born in 1995; Sananas and Georgia Horackova are born in 1989. They created YouTube blogs between 2010 and 2011 and have between 2-3 million (Sananas and EnjoyPhoenix) to 300-800 thousand followers (Dazzlingdrew and Georgia Horackova) on YouTube. On Instagram, EnjoyPhoenix, Sananas, GeorgiaSecret and Dazzlingdrew count 4,3 million, 1,8 million, 289 thousand and 150 thousand followers respectively. None of the four influencers are traditional celebrities or official experts on beauty and fashion.

79% of the sample is composed of users who are representative of the Z Generation (born after 1995), 17% of Y generation (born between 1980 and 1995), 4% belong to X Generation (born between 1960 and 1980). We have excluded from our study sample baby boomers born before 1960 (less than 1% of the original sample). There was no data collected from the earlier generations in our study.

A majority of respondents are female that is not surprising as beauty and fashion industry are mostly related to female field of interest. Less than 1% of the responses were from male and these were excluded from the study sample. The final sample included 1209 exclusively female responses, distributed as follows: 304 for EnjoyPhoenix; 304 for Sananas; 303 for Georgia Secret and 298 for Dazzlingdrew.

4.2. Statistical procedures

A partial least squares (PLS) approach with SmartPLS software 3.2.7 (Ringle and Becker, 2015) was used to test the model. PLS is able to model latent variables without requiring strict normal distributions of data (Ringle et al., 2012). This approach is appropriate to this study whose aim is the identification of the key drivers of constructs and not theory confirmation that is best handled by covariance-based structural modeling. We started with a confirmatory phase to assess the measurement model.

We used factor analysis to verify the validity of all constructs. The results in table 2 show the factor loading for all measures is above 0.8 or above 0.9 allowing us to keep all measures. The internal consistency of variables is verified using Cronbach’s Alpha, the composite reliability is also verified. The results are displayed in table 1 showing high internal consistency. Average variance extracted (AVE) is higher than 0.5 that is adequate for convergent validity according to Hair et al. (2014).

To address discriminant validity, we first compare Average Variance Extracted (AVE) and Shared Variance between variables as suggested by Fornell and Larcker (1981). The AVE of the latent constructs should be greater than the squared correlation between variables. Table 3 represents the related results where all of the square root of the AVE (highlighted in bold) are greater than the correlations between variable that confirms the discriminant validity of the constructs. We also applied heterotrait-monotrait ratio of correlations considered to be more reliable to further confirm the discriminant validity (Henseler et al., 2015). Table 4 shows satisfactory results where all values are below 0.9. We observe that some constructs, such as credibility, are correlated with other constructs that suggests it shares some of the predictive power over the dependent variables with them (Farrell, 2010).

Following the validation and reliability verification, we applied bootstrapping analysis with 5000 re-samples to the whole sample to examine the structural validity of the model (hypotheses testing). Then, we applied two multi-group analyses (MGA). First, we control the age variable and investigate whether the generational effect is significant in this study. The second MGA is applied in order to test our hypotheses for each of the 4 influencers we are investigating, and then to compare the results.

Table 1: Latent variables’ reliability (CA = Cronbach’s α ; CR = Composite reliability; AVE = Average Variance Extracted)

Latent variables	CA	CR	AVE
Social Attractiveness	0,944	0,973	0,947
Physical Attractiveness	0,915	0,959	0,922
Attitude Homophily	0,932	0,967	0,936
Para-social Interaction	0,768	0,893	0,807
Credibility	0,915	0,937	0,747
Intention to Purchase	0,848	0,929	0,867

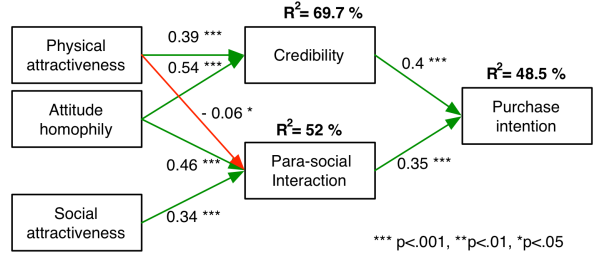


Figure 2: Findings for the global sample (N=1209). Accepted hypotheses are highlighted in green.

5. Results

First, we performed the analysis on the global sample. Table 5 shows path coefficients and significance of the structural model for the whole sample and figure 2 shows the results visually. With regard to the levels of R^2 , the model explains 48.5% of the variance of the intention to purchase construct, 52% of the variance of PSI and 69.7% of the variance of credibility, which is very satisfactory in social sciences (Chin, 1998).

Table 6 and figure 2 show the hypotheses testing results. All our hypotheses are supported, except H5 (physical attractiveness $\rightarrow$ para-social interaction) for which we have a significant path (p -value = 0.04) but it is unexpectedly negative rather than positive ($\beta = -0.06$), highlighting that physical attractiveness is negatively related to para-social interaction. However, as expected, physical attractiveness is positively related to the credibility of the blogger ($\beta = 0.39$).

Intention to purchase is determined by both para-social interaction and credibility where the impact of credibility is slightly higher than PSI ($\beta = 0.4$ and $\beta = 0.35$ respectively). PSI is positively determined by attitude homophily and social attractiveness, where social attractiveness has a slightly higher impact on PSI ($\beta = 0.46$ comparing to $\beta = 0.34$).

Table 2: Construct measures validity. Loading values are all significant at $p < 0.01$.

Latent variables with indicators	Mean	Sd	Loading
Social Attractiveness			
I think she could be a friend of mine (Q50)	3.42	1.133	0.974
I would like to have a friendly chat with her (Q60)	3.52	1.123	0.972
Physical Attractiveness			
I think she is quite pretty (Q70)	3.96	0.973	0.963
I find her very attractive physically (Q80)	3.72	0.972	0.957
Attitude Homophily			
She thinks like me (Q90)	3.14	0.998	0.965
She shares my values (Q100)	3.23	1.022	0.970
Para-social Interaction			
I look forward to watching her videos. reading her posts and comments (Q200)	3.20	1.238	0.934
I would follow her and interact with her on other social networking sites (Q300)	2.74	1.295	0.861
Credibility			
I find this blogger expert in her domain (Q400)	3.48	1.036	0.869
I find this blogger efficient in her job (Q700)	3.08	1.108	0.848
I find this blogger trustworthy (Q600)	3.48	1.064	0.916
I think this blogger cares about her followers (Q500)	3.78	1.037	0.825
This blogger updates regularly her content (Q800)	3.54	1.054	0.871
Intention to Purchase			
I would purchase the products promoted by the blogger in the future (Q900)	2.99	1.287	0.921
I would encourage people close to me to buy the products promoted by the blogger (Q1000)	2.79	1.168	0.941

Table 3: Discriminant validity (Fornell and Larcker, 1981). The diagonal elements in bold represent the square root of the average variance extracted for the respective construct.

	Correlations					
	1	2	3	4	5	6
1. Purchase Intention	0,931					
2. Para-social Interaction	0,633	0,898				
3. Physical Attractiveness	0,379	0,435	0,96			
4. Social Attractiveness	0,521	0,666	0,655	0,973		
5. Credibility	0,659	0,708	0,706	0,691	0,864	
6. Attitude homophily	0,579	0,694	0,587	0,69	0,673	0,968

Table 4: Heterotrait-Monotrait Ratio (HTMT) (Henseler et al., 2015)

	1	2	3	4	5
1. Purchase Intention					
2. Para-social Interaction	0,754				
3. Physical Attractiveness	0,431	0,49			
4. Social Attractiveness	0,58	0,76	0,705		
5. Credibility	0,732	0,809	0,78	0,85	
6. Attitude homophily	0,648	0,805	0,636	0,842	0,833

Table 5: Structural model evaluation (Global sample N=1209). f^2 = effect size of path coefficient; R^2 = coefficient of determinants; Q^2 = predictive relevance/Stone-Geisser Q^2 (blindfolding procedure with omission distance of 7)

Construct/structural path	f^2	R^2	Q^2
Social Attractiveness			
Social attractiveness → Para-social interaction	0.079		
Physical Attractiveness			
Physical Attractiveness → Para-social interaction	0.040		
Physical attractiveness → Credibility	0.336		
Attitude Homophily			
Attitude homophily → Para-social-interaction	0.160		
Attitude homophily → Credibility	0.634		
Credibility			
Credibility → Purchase intention	0.164	0.697	0.672
Para-social interaction			
Para-social interaction → Purchase intention	0.118	0.520	0.450
Purchase intention			
		0.485	0.177

For further investigation, we performed multi-group analysis by respondent’s age and by influencer; the results are displayed in tables 7 and 8 respectively. The results are globally consistent with the full sample analysis. H3 (physical attractiveness → para-social interaction) is rejected for all cases in both tests mostly because the statistical results are non significant: the null-hypothesis cannot be rejected. The path is negative and significant only for one influencer Georgia Secretsyst ($\beta = -0.18$).

Analysis by generation shows a insignificant path for H4 (social attractiveness → para-social interaction) for the generation Z (the youngest in our sample). The same hypothesis is not supported for Georgia Secret.

6. Discussion

Both credibility and parasocial-interaction with bloggers impact the purchase intention of followers, meaning that a follower goes through both affective and internalization processes when influenced to purchase a product. Global sample analysis highlights a slightly stronger impact of credibility on purchase intention. However, a closer look shows that the impact of para-social interaction is slightly stronger than credibility for three out of four influencers and much stronger for the generation Z. It means that followers who are attached, or even addicted, to the blogger are more likely to purchase than less attached followers. This is particularly true for the younger generation: the credibility of the blogger has a positive impact on purchase intention, while having a much more silent role than

PSI ($\beta = 0.7$ comparing to the $\beta = 0.3$ for the generation Z). Generation Y shows similar results, while the credibility becomes predominant for the older generation.

Homophily affects both para-social interaction and credibility, showing that value sharing is a strong aspect of persuasion. Creating an affective link with a blogger comes through shared values and those shared values are also indicative of the quality of the content the blogger proposes. The shared values are related to the internalization process that seems to predominate in our results.

Interestingly, for younger generations, para-social interaction in the case of beauty influencers and their followers is more like an addiction than a friendly relationship. We were not able to confirm a positive relationship between social attractiveness and PSI for generation Z, meaning that followers would not necessarily feel a friendship-like relationship with a blogger they regularly follow, as was supposed by Berryman and Kavka (2017). That is not the case for the older generations (Y and X) where this path is significant. This could indicate a generational switch in the conception of para-social interaction, which becomes more like an addiction and content consumption than an illusory relationship (Moghavvemi et al., 2017; Romo et al., 2017).

Shared values are certainly important for the youngest respondents when they decide to regularly follow a blogger, unlike the attractiveness (social or physical) of the presenter. Even considering beauty and fashion bloggers, the followers would not be more attracted to a more beautiful blogger.

Table 6: Results of hypothesis testing (Global sample N=1209). One-tailed test.

	Hypothesis/Structural path	β	t-value	p-value	Result
H1	Credibility $\rightarrow$ Purchase intention	0.409	14.202	0.000	Accepted
H2	PSI $\rightarrow$ Purchase intention	0.346	10.655	0.000	Accepted
H3	Physical attractiveness $\rightarrow$ PSI	-0.059	2.096	0.039	Rejected
H4	Social Attractiveness $\rightarrow$ PSI	0.344	8.433	0.000	Accepted
H5	Attitude homophily $\rightarrow$ PSI	0.457	12.596	0.000	Accepted
H6	Physical attractiveness $\rightarrow$ Credibility	0.394	14.291	0.000	Accepted
H7	Attitude homophily $\rightarrow$ Credibility	0.541	20.623	0.000	Accepted

Table 7: Results of hypothesis testing (Control Variable: Age). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. A = accepted; R = rejected. One-tailed test.

GenZ				GenY			GenX		
Hypothesis	β	t-value	Result	β	t-value	Result	β	t-value	Result
H1	0.279	2.142*	A	0.405	7.847***	A	0.405	7.847***	A
H2	0.697	6.715***	A	0.549	11.201***	A	0.309	8.240***	A
H3	-0.061	0.279	R	-0.075	0.918	R	-0.058	1.875	R
H4	0.030	0.134	R	0.289	2.496*	A	0.356	7.806***	A
H5	0.518	2.344*	A	0.515	4.793***	A	0.448	4.793***	A
H6	0.431	3.935***	A	0.396	5.850***	A	0.391	12.342***	A
H7	0.491	4.529***	A	0.566	4.532***	A	0.536	18.362***	A

This way, online social beauty bloggers are different from traditional celebrities and luxury influencers, as the relationship between physical attractiveness and PSI is not confirmed in the majority of tests in our study (Erdogan, 1999; Lee and Watkins, 2016).

However, multi-group analyses by influencer confirm that social attractiveness is positively related to the para-social interaction for three out of four influencers: DazzlingDrew, EnjoyPhoenix and Sananas. Georgia Secret differs from the other beauty influencers because her content focuses mostly on fitness and yoga, not covered by other three. Georgia Horackova's YouTube channel provides multiple fitness tutorials, and her Instagram feed features fitness-related clothes. She also publishes fashion-related content like look-books and outfits, but does not publish makeup tutorials and rarely features makeup or beauty products. Additionally, she is the only blogger having a significant path between physical attractiveness and PSI. The relationship is negative ($\beta = -0.184$) suggesting the more the user finds her beautiful, the less he/she is attached to her. The relationship between PSI and purchase intention is stronger for her than for the other influencers ($\beta = 0.635$) and the relationship between credibility and purchase intention is the weakest ($\beta = 0.248$). That is interesting, because she positions herself as a yoga instructor

(suggesting she is an expert). Thus, her followers probably look at Georgia's photos and videos for the content and not particularly for the speaker and they purchase the products when they are attracted to the content that could be pleasing or useful.

Unlike Lee and Watkins (2016), who found that physical attractiveness is related to PSI in the case of Youtube and luxury products, we found no, or a negative effect of physical attractiveness on PSI for beauty bloggers that are not particularly positioned as "luxury". This counter-intuitive result concerning could, in fact, be explained by the role played by bloggers and influencers on social media, which appears to be different from those played by models in traditional media, such as TV and magazines particularly in the luxury market. Models generally exhibit a perfect physical appearance and seldom 'talk' to people directly, explain their preferences or provide useful day-to-day tips. This is, on the contrary, exactly what bloggers do on social media. They are therefore perceived as more authentic, as observed by Djafarova and Rushworth (2017). This is why attitude homophily increases PSI as expected, but not physical attractiveness. Thus, physical attractiveness can be less important for customers seeking mass-market beauty and fashion products than for the luxury market where they are more likely seeking higher quality.

Table 8: Results of hypothesis testing (Control Variable: blogger). A = accepted; R = rejected. One-tailed test. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Hypothesis	DazzlingDrew			EnjoyPhoenix			Georgia Secret			Sananas		
	β	t-value	Result	β	t-value	Result	β	t-value	Result	β	t-value	Result
H1	0.363	5.525***	A	0.338	8.228***	A	0.248	3.957***	A	0.395	9.353***	A
H2	0.223	3.373**	A	0.545	12.371***	A	0.635	10.278***	A	0.532	13.118***	A
H3	0.031	0.523	R	-0.051	0.770	R	-0.184	2.974*	R	-0.060	1.064	R
H4	0.339	5.598***	A	0.228	3.251**	A	0.194	1.457	R	0.239	2.638*	A
H5	0.419	8.192***	A	0.474	7.514***	A	0.657	4.991***	A	0.371	5.617***	A
H6	0.263	5.221***	A	0.306	6.614***	A	0.329	4.375***	A	0.458	8.786***	A
H7	0.501	11.617***	A	0.543	11.858***	A	0.642	8.580***	A	0.509	12.433***	A

Another explanation of this result could be in the complex and subjective definition of what is 'beautiful'. Beauty could be related to the body, the face, the hair or even the voice and how all these are perceived (or magnified) on social media. Also, beauty bloggers can wear different makeup, from natural to extravagant, that would be perceived differently. For example, Sananas always wears strong makeup, while DazzlingDrew and EnjoyPhoenix mostly feature 'natural' and less visible makeup. Georgia Secret opts for a makeup that is more intense than DazzlingDrew but more "every-day" than Sananas. A negative relationship between physical attractiveness and PSI for Georgia can be explained by the fact that the makeup she always wears could be considered beautiful but irrelevant to fitness. Also, some influencers show their faces without makeup, with natural imperfections that could make them seem "less beautiful" i.e. less perfect, but more "natural" and similar to normal people.

Finally, the "beauty" could be related to the clothes a blogger wears and features in photos and videos. Followers would not necessarily like all the outfits presented by the influencer, or consider them beautiful, but could be generally attracted by the style of the person, which would make them look for other posts and outfits. Then, the purchase intention could be affected by a similarity of a style and not by the beauty per se of the blogger (Kulmala et al., 2013).

6.1. Theoretical contribution

Our study adds to knowledge on the recent research stream of influencer marketing by investigating the combined effects of the influencers perceived credibility and parasocial interaction on the purchase intention of users of social media platforms. It is one of the first studies in this topic using a large data set (1209 respondents) collected from the audiences of four popular influencers in the beauty and fashion sector in France. To explain the online influence mechanisms, we have applied theories originally developed for offline interactions, such as persuasion theory and para-social interaction theory, and discussed to what extent they help to enlighten the influential relationships between an online content creator (who is not necessarily a deemed celebrity outside social media context) and his or her audience.

This work contributes to the existing literature on the generational effect on social media interactions (Goodrich, 2013; Muk, 2013), and how they

have to be taken into consideration in influencer and social media marketing. Our study pointed out, for example, that younger generations value para-social interaction and what they perceive as their personal attachment to the influencer more than the perceived credibility. While this attachment is more related, for this category of users, to homophily than it is to social or physical attractiveness. This means that new canons of influence are experienced by younger generations, especially generation Z, that need a more thorough examination, since many studies are currently focusing on Millennials.

6.2. Practical implications

The results of this study could be beneficial for brands to help them establish affiliate relationships with influencers according to their persuasive and para-social interaction talents. It is important for brands to understand the persuasive cues activated by the influences in order to define a fine balance between creativity and control. It is also beneficial to bloggers who wish to adopt more persuasive strategies while creating the content and collaborating with brands.

Both credibility and PSI are positively related to the purchase intention, meaning that the longtime followers of trustworthy influencers who care about their followers and show expertise on the subject are more likely to purchase the featured products. For the younger audience, a connection making them look for further information posted by their favorite blogger is more persuasive than credibility, confirming the importance of creating such a link. Thus, a brand targeting young population would find potential customers among the most attached or even addicted followers. As for bloggers, they should not only work on community growth, but should also care about already acquired followers to build strong para-social relationships. Being socially attractive and friendly can help to reach generations Y and X. For the younger generation, the para-social relationship would more likely be related to an 'addictive' content.

The perceived similarity with an influencer in terms of shared values is strongly related to both credibility and PSI. Brands should understand the values that are conveyed by the influencer and also valued by the potential customers, as the bloggers would be more influential on followers that think like them (for example, values related to the eco-behaviour Dwivedi et al. (2015)). Therefore, the

branded content should be related to the same values. Influencers can use this insight and adopt attractive, common or trending values of the digital population. Thus, trending values can be shared with a larger audience while building both a para-social relationship with followers and enhancing the credibility of the blogger. However, a significant change in values may have the opposite effect.

The (perceived) expertise on the subject and the goodwill remain important for credibility and persuasion and should be transmitted via the created content, particularly while targeting the older generation. While physical attractiveness is related to credibility of beauty and fashion bloggers, seeking a perfect physical appearance is not the ultimate goal of the followers.

7. Conclusion

Nowadays, companies are moving toward a more digitized communication to promote their products and brands, where social media and influencer marketing become unavoidable when defining marketing strategies.

In this paper, we study how credibility and para-social interaction are related to the purchase intention. Additionally, we study the indirect relationship of attitude homophily, physical attractiveness and social attractiveness with the purchase intention. The results of this study could be beneficial for influencers helping them to adopt more persuasive behavior. It is also important for brands to understand the persuasive cues related to the influencers who present their products.

From a theoretical point of view, our results suggest the need for further studies on para-social interaction, that may need re-conceptualization regarding online influencers. We have not found any evidence that physical attractiveness are related to PSI, particularly for young populations that seem not to value social attractiveness either, probably meaning that the concept of PSI has changed. More research could uncover what drives an audience to continue following the blogger and what attaches the younger generation to an influencer.

Our study has some limitations related to its specific focus on a female audience located in France in a single sector: beauty and fashion. This could restrict the generalization of the results to other audiences and other sectors. Influencers we study in this work are present and popular on multiple

platforms, such as YouTube and Instagram, commonly posting different content. Thus, our respondents could have been exposed to different types of content coming from the two platforms. Future work could investigate the persuasion clues for each platform separately considering their content specificity. Other social media platforms specializing in more ephemeral interactions, such as Snapchat, could be further studied to compare the results.

It could also be interesting to further study the concept of beauty and its different facets in the case of digital influence. Thus, not only body and face, but voice, style, lifestyle, make-up or other more specific aspects of beauty could be relevant to the persuasion.

This study may also have been limited by the hypothetico-deductive research design adopted. We used a survey-based approach and adapted items defined from the literature to measure psychometric variables related to a complex and evolving psychosociological phenomenon. Future studies could address the subject using qualitative or mixed research designs to explore more emergent insights. Our study can be extended by a content analysis that could uncover arguments and content-related persuasion cues. The interaction between content generated by an influencer and comments and reactions from the followers could be further investigated.

Influence encompasses dynamic and ubiquitous processes that continue to emerge and evolve, depending on the influencers, audiences, brands and social media channels. More research is though certainly needed to bring light on this new and complex phenomenon.

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